Finite Quantum Mechanics

Can Quantum Mechanics Be Presented on a Finite Carrier — and If Not, Exactly Where Does It Break?

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Abstract

We study finite-dimensional quantum models over specified rational and algebraic scalar fields, distinguishing a finite presentation from a finite set of states. The principal results are exact constructions and obstructions under explicit assumptions on states, effects, composition, dynamics, and readout. A normalized clock seed gives a constructive finite Weyl intertwiner. Equal population reports can fail to preserve answers to a common future experiment. An unconditional measure-and-prepare transfer of an unknown qubit has average survival probability at most three quarters on four specified preparations, with an attaining strategy. Finite Bell tables distinguish support compatibility from the existence of a positive common joint distribution, while central-sector weights constrain purification in a specified direct-sum tensor model.

Exact finite examples, written arguments, and selected formalizations accompany one another with separate scopes. A positive global joint law implies the CHSH bound

$|S| \leq 2$; a supplied rational quantum model reaches $S = 14/5$; and a supplied tensor-state-observable model over $\mathbb{Q}(\sqrt{2})$ attains $S = 2\sqrt{2}$ and has an exact sum-of-squares upper certificate. These results do not derive the quantum state space, Born pairing, tensor composition, or continuous dynamics from primitive difference. They identify which supplied structures each conclusion uses and which reconstruction questions remain open.

Guide to the paper

The main text is organized by mathematical questions. Source provenance and reproduction details accompany the results without determining their truth.

Question	Location	Evidential boundary
Which finite and algebraic structures are being assumed?	§1	Declared carrier and scalar data are not derived conclusions
What counts as a proof or finite certificate?	§§2–3	Evidence is matched to the quantifiers of the statement
Which mathematical results are established?	§§4 and 11	Each theorem keeps its scalar, dimension, positivity, and composition premises
Where does the proposed reconstruction remain open?	§§5, 7, and 9	A missing selection argument is not an impossibility theorem
How can the computations be reproduced?	§§6, 10, and 12	Validation and serialization checks do not establish physical predicates that fields only name
What do the carrier and measurement examples establish?	§13	Conditional operational separations; no individual-outcome selection law

1. Finite presentations, supplied structure, and target examples

A finite-dimensional quantum model need not have finitely many states. This paper distinguishes four notions:

1. a finite graph or cell complex;
2. a finite presentation by exact algebraic data;
3. a finite-dimensional vector space, which normally has infinitely many normalized states; and
4. a finite list of preparations, effects, or experiments.

On a declared directed graph the primitive coboundary is

$$(d_0 f)(u, v) = f(v) - f(u).$$

For an edge cochain x, a weighted quotient such as

$$\Phi_1([x]) = \inf_{\alpha} \|x - d_0 \alpha\|_{1,w}$$

measures the residual after the admitted vertex repairs. This construction is mathematically useful, but it does not by itself select complex scalars, positivity, inner products, tensor composition, the Born pairing, or continuous time. Those choices must be supplied or derived separately.

1.1 An eight-rung inventory

The eight rungs below are an assumption audit, not an eight-step derivation of quantum mechanics.

Rung	Established once the inputs are supplied	Additional supplied structure	Remaining boundary
1. Difference	Finite differences, equality tests, and graph coboundaries	A carrier and equality relation	No preferred norm, topology, or physical geometry is selected
2. Quotient repair	Exact finite quotient values and primal/dual certificates	Coefficient space, weights, and repair image	The quotient does not choose a quantum state space
3. State–effect pairing	Positive normalized weights and conditional identities in stated models	Positive states, effects, and a pairing	The Born rule is not derived from difference
4. Phase	Relative phases and phase-sensitive interference on a supplied complex or cyclotomic carrier	Complex conjugation and a phase group	Population data alone do not select the phase fiber
5. Composition	Product states, entangled states, partial traces, and Bell tables in a supplied tensor model	Tensor composition and positivity	Tensor products are not forced by graph products

Rung	Established once the inputs are supplied	Additional supplied structure	Remaining boundary
6. Reversible dynamics	Exact finite unitaries and rational Cayley steps	An inner product, reversible maps, and a time step	Continuous time and a generator require topology and interpolation
7. Measurement and channels	POVM probabilities, conditional updates, exact channels, and finite dilations	Effect algebra, instruments, and completely positive maps	Unitary dilation does not select one realized outcome
8. Representation boundary	Finite frame constraints, contextuality witnesses, and finite-dimensional commutator obstructions	The hypotheses of Gleason/Busch-type representation theorems	Restricted algebraic ray sets do not automatically extend to complete projective geometry

Two elementary cautions are important. First, the difference and equality relation do not select the norm, topology, or physical geometry needed later; they do not exclude every metric construction. Second, projective state space is

$$\mathbb{P}(\mathcal{H}) = (\mathcal{H} \setminus \{0\})/\mathbb{C}^\times \cong S(\mathcal{H})/U(1),$$

where $S(\mathcal{H})$ is the unit sphere. The group quotient $\mathbb{C}^\times/U(1)$ instead records positive scale and is not the projectivization of the state space.

1.2 Twenty-two finite targets

The target list tests exact finite presentations against familiar structures. Each row states the object actually considered; it is not a claim that the corresponding general physical theory has been reconstructed.

Target	Finite object and present scope
1	Primitive difference and zero self-difference on a declared carrier
2	A diamond-shaped 2-complex with exact cyclotomic holonomy data
3	A three-path amplitude fixture with vanishing Sorkin third-order term [7] under a supplied quadratic readout
4	Rational Cayley transforms in $\mathfrak{so}(8, \mathbb{Q})$ and exact orthogonality checks

Target	Finite object and present scope
5	Rational S_3 braid relations and recoupling-shaped examples; no general pentagon theorem
6	Cabello's 18-ray parity system, its 146-vertex rational frame polytope, and exact positivity filtering
7	Signed permutation unitaries on a finite layered model
8	A rational CHSH model with $S = 14/5$ and an exact positive-extension obstruction
9	The three-qubit bit-flip code, whose distance is one against unrestricted Pauli errors
10	Four-branch teleportation identities over exact Gaussian-rational data
11	Superdense-coding identities on a supplied Bell pair and gate set
12	A four-qubit $[[4, 2, 2]]$ stabilizer example with exact syndrome and distance checks
13	Prime-5 discrete Wigner phase-point operators and selected covariance/ negativity witnesses
14	The Shor $[[9, 1, 3]]$ stabilizer construction and single-qubit error syndromes
15	Four-point phase-estimation identities over a specified cyclotomic carrier
16	A finite toric/surface-code incidence model and its declared anyon relations
17	Four-item Grover search as two exact Householder reflections
18	Finite Clifford conjugation identities and a scoped transversal-gate limitation [25]
19	The finite commutator trace obstruction and a constructive finite Weyl representation
20	Exact Richardson weights and the variance cost of finite zero-noise extrapolation
21	A finite quantum walk on a supplied Cayley graph with exact norm preservation
22	Exact mutually unbiased-basis tomography in dimensions three and four, with positivity checked separately from linear reconstruction

The detailed target definitions, implementation addresses, and formalization scope are in the source-notes supplement. Three results deserve emphasis here: the Bell separation in §11.4, the Cabello census in §11.5, and the following finite Weyl construction.

1.3 Constructive finite Weyl equivalence from an anchored seed

Let $N \geq 1$ and let $K \subseteq \mathbb{C}$ be a conjugation-stable subfield containing a primitive Nth root of unity ω. Equip K^N with the standard Hermitian pairing. Supply unitary $N \times N$ matrices X', Z' over K satisfying the finite Weyl relations [4]

$$Z'X' = \omega X'Z', \qquad (X')^N = I,$$

and a normalized vector $v \in K^N$ satisfying

$$Z'v = v, \qquad v^\dagger v = 1.$$

Define the canonical shift and clock by $Xe_j = e_{j+1 \pmod N}$ and $Ze_j = \omega^j e_j$, for $0 \leq j < N$. Then

$$W = \begin{pmatrix} v & X'v & \cdots & (X')^{N-1}v \end{pmatrix}$$

is unitary and satisfies $X'W = WX$ and $Z'W = WZ$. The joint commutant of X' and Z' consists of scalar matrices. Any two unitary intertwiners between the fixed pairs differ by multiplication by a unit-modulus scalar in K.

Proof. Induction in the Weyl relation gives $Z'(X')^j v = \omega^j (X')^j v$. These N eigenvalues are distinct. Since Z' is unitary, their eigenvectors are orthogonal, and unitarity of X' makes all N orbit vectors have norm one. The explicitly N-dimensional carrier makes them a basis. The period of X' closes the shift relation, and the displayed eigenvalue equation gives the clock relation. A matrix commuting with the canonical clock is diagonal; commuting with the shift makes all its diagonal entries equal. Conjugating by W establishes the commutant statement, and applying it to the ratio of two intertwiners establishes uniqueness up to scalar.

The normalized eigenvalue-one seed and scalar field are supplied inputs. This result does not find or normalize such a seed over every number field. If a supplied seed has eigenvalue ω^r instead, replace it by $(X')^{-r}v$ before using this construction. On a larger carrier the orbit spans a cyclic invariant subspace, not necessarily the whole carrier; no scalar-commutant conclusion for the larger carrier follows.

2. Standards of evidence

The mathematical statement, its premises, its implementation and its reproduction record are separate. We use the following evidence classes throughout the paper:

Class	What it supports	What it does not support
Written proof	The quantified statement under the assumptions used in its argument	A claim of Lean verification or physical realization
Scoped Lean proof	The actual elaborated declaration, including its input hypotheses and coefficient types	A broader theorem suggested by a filename, comment or surrounding prose
Exact finite computation	An identity, enumeration or certificate replay on its declared input domain	An unrestricted theorem from a sample of instances
Numerical demonstration	A reported approximation with its algorithm and tolerance	Exact zero, exact positivity or exact optimality without additional certificates
Imported theorem or conditional model	A standard theorem with its hypotheses, or a result conditional on supplied structures	Derivation of those hypotheses or structures from finite history data
Historical observation / unreproduced claim	What an identified earlier source or execution record reported	A fresh successful run or endorsement of every present source claim

A finite computation can decide a finite theorem completely, and a checked linear basis can establish an identity for its entire finite-dimensional space. Conversely, a general written proof remains a proof even when no Lean mechanization is supplied. Neither direction should be collapsed into the ledger's software status labels.

The most important conclusions have these boundaries:

Conclusion	Premises and supporting argument	Current scope
Finite Weyl equivalence and scalar commutant	Primitive root, unitary $N \times N$ pair, period and normalized eigenvalue-one clock seed; explicit eigenvector-orbit proof in Target 19	Written theorem and exact verifier; Lean checks the stated commutant fragment
Born normalization	Supplied positive trace-one state, POVM summing to identity and trace pairing; linearity of trace	General algebraic identity; specialized two-level Lean models and exact channel checks

Conclusion	Premises and supporting argument	Current scope
Gleason/Busch representation [1,3]	Full projective domain at dimension at least three, or an all-effects additive positive normalized domain	Imported theorems; rational-plane obstruction and finite tomography do not prove a general number-field representation theorem
Phase-blind reports can lose future answers	Two equal present reports separated by an admitted common future experiment	Explicit exact counterexample; no exclusion of every classical enlarged model
Bell joint-extension obstruction	Four supplied pair tables and one positive joint law on all sixteen local assignments	Finite convex separation and CHSH proof; a physical Bell reading also assumes setting independence and local response factorization
Classical-storage survival at most $3/4$	Full qubit positive states/effects, affine unconditional encoding/retrieval, unknown input and no quantum side memory	General written proof with sharp control in §13.7; exact native instances
Central-sector purification obstruction and bounds	Full states of direct-sum complex matrix algebras and standard tensor composition	General written proof in §13.8; sharpness needs sufficient ancilla capacity; Lean assumes the pure-block representation
Native history quotient	A declared finite reachable history/action system and protected future questions, including availability	Minimal report relative to those questions; no quantum scalar, positivity or composition selection

Repository-only checks validate status vocabulary, evidence paths and named Rust test functions. They enforce bookkeeping; they do not decide whether a file establishes a claim. The source-by-source discussion in the source-notes supplement (its §4, subsections 11.1–11.60) supplies that interpretation and retains counterexamples and open obligations. Its old execution counts are historical unless an explicitly dated current replay is named.

3. Claims and supporting evidence

The evidence obligation is determined by the statement's quantifiers. A general mathematical statement requires an adequate written proof or a formal proof of the same statement. A claim over a finite domain may instead be settled by a complete exact

enumeration, provided the domain and enumeration are justified. An example test establishes only the stated example.

Accordingly, each unresolved statement must receive one of the following treatments:

1. an adequate written proof;
2. a matching formal proof;
3. a complete exact finite argument appropriate to its quantifiers;
4. a restriction to the domain actually established;
5. an explicit statement that it remains open; or
6. withdrawal after a counterexample.

File existence, compilation, schema validation, serialization, access control, and a change of status label cannot decide whether this obligation has been discharged. The repository maintains a historical evidence register for reproduction, but its row counts and labels are not a measure of mathematical progress and are not part of this monograph's scientific summary.

The sixty-seven source manuscripts are preserved in the paper archive. Six are redirect, alias or annex stubs whose mathematical content is treated with the destination result: the residual-density redirect (source-notes supplement §11.4), the finite-realization redirect (§11.6), the earlier phase-selector alias (§11.27), and the annex stubs for phase-gauge residuals (§§11.28–11.29), the no-signaling model (§§11.33–11.34) and the completely positive import boundary (§11.38).

4. Established results

The results below are stated by mathematical content rather than by repository status. Their detailed hypotheses, arguments, and source addresses appear in §§11–13.

1. **Finite Weyl equivalence from an anchored seed (§1.3, Target 19).** Given the stated conjugation-stable scalar field $K \subseteq \mathbb{C}$, primitive Nth root ω, unitary $N \times N$ pair X', Z' with $(X')^N = I$, and normalized clock eigenvector $Z'v = v$ with $v^\dagger v = 1$, the orbit columns form an orthonormal eigenbasis and give an explicit unitary intertwiner W. The scalar-commutant conclusion on K^N follows by the displayed basis argument. Finding or normalizing such a seed over an arbitrary number field is not included; on a larger carrier the orbit spans an invariant cyclic subspace.
2. **Finite cochain certificates (source-notes supplement §§11.3, 11.4 and 11.8).** An exact potential establishes membership in $\operatorname{im} d_0$; a separating functional vanishing on that image establishes non-exactness; and a nonzero $d_1 x$ establishes failure of closure. Numerical linear programming is reported separately.
3. **Power-family context dependence (source-notes supplement §11.58, positive-power context independence).** Two rational contexts assign a shared ray the weights $1/2$ and $1/(1 + (3/5)^p + (4/5)^p)$. For positive real p, equality holds exactly at $p = 2$. This classifies the specified family, not all frame functions.
4. **Positive Bell extension (§13.8).** The full-support rational table has CHSH value $14/5$. Every deterministic assignment is support-compatible, yet no nonnegative common joint distribution reproduces the four marginals. The separating

inequality is verified on all sixteen deterministic vertices.

5. **Classical-storage survival bound (§13.7).** Under the stated full qubit state/effect domain, affine unconditional measure-and-prepare transfer, unknown input, and absence of quantum side memory, the average survival on four preparations is at most $3/4$, and the bound is attained.
6. **Central-sector bounds (§13.8).** In the specified direct-sum matrix-algebra state space with standard tensor composition, central weights bound extension purity and the trace distance from a pure-composite marginal. Attainability requires sufficient ancilla capacity.
7. **Primal and dual trace-norm certificates (source-notes supplement §11.35).** For $H = P - N$, positivity of P, N, the inequalities $-I \preceq W \preceq I$, and

$$\operatorname{tr}(WH) = \operatorname{tr} P + \operatorname{tr} N$$

certify the claimed value independently of the implementation's label.

8. **Phase-blind descent with a retained zero point (§11.1).** When the quotient keeps the zero vector as a distinct output, the population report of Ac depends only on that of c exactly when every row of A has at most one nonzero entry. On projective space alone the rank-one all-ones matrix is a counterexample. Written proof and exact two-dimensional replay.
9. **Trace-zero repair duality (§11.2).** The dual norm of $\frac{1}{2}\|\cdot\|_1$ on $\mathrm{Herm}_0(n)$ is the spectral diameter. On $K_{2,3}$ with the stated qutrit defects the trace-zero repair value is 6 and the all-Hermitian value is $9/2$, each certified by an exact primal potential and a dual witness [13].
10. **CPTP monotonicity selects the trace norm (§11.3).** A family of norms on trace-zero Hermitian matrices that no CPTP map increases is one positive multiple of the trace norm. Written proof by exact measure-and-prepare transport.
11. **The Cabello frame polytope (§11.5).** The frame polytope of the eighteen Cabello rays has exactly 146 vertices, all half-integral, and none of their unique trace-one real-symmetric representatives is positive semidefinite. Exact enumeration with characteristic-polynomial certificates.

5. Open problems and limitations

The unresolved scientific questions are stated here directly rather than by internal tracking codes. A question is closed only by a dated argument or replayable result; renaming or reclassifying it does not resolve it.

The seven principal finite-to-continuum boundaries are:

1. **The continuum Hilbert lift.** No source constructs the passage from a finite presented carrier to an infinite-dimensional Hilbert space. Source Notes §11.52 explicitly refuses that conclusion.
2. **Born uniqueness (Gleason [1]).** The Born rule is imported as a declared input throughout, never derived.
3. **Continuous time and Stone's theorem.** The finite examples distinguish finite-order integral evolutions from specified infinite-order rational rotations. Rational unitaries can also have finite order (the identity and rational quarter-turn suffice). A dense cyclic orbit has a closure only after choosing an ambient topology;

that closure neither selects a time parameter nor a unique logarithm. Topology, continuous interpolation and its generator therefore remain additional inputs in both sectors.

4. **The Gleason dimension condition at $n \geq 3$.** Gleason's theorem [1] uses every orthonormal basis of a space of dimension at least three, and Busch's version [3] uses every effect. A finite ray set gives only part of that conclusion. On the Cabello set in $\mathbb{Q}^4$, on its three- and six-basis prefixes, and on the 24-ray Peres set, every function that sums to one on each declared basis is the restriction of a real-symmetric trace-one form (exact rank computation). The form is unique once the ray projectors span the ten-dimensional space of real symmetric 4×4 matrices, which holds on every listed set except the three-basis prefix. Nonnegativity on the declared rays does not give positive semidefiniteness: the 146 vertex representatives of §11.5 are nonnegative on the eighteen Cabello rays, the research note `research/qm_210_frame_function_finite_map.md` displays a rational trace-one form nonnegative on both the Cabello and the Peres rays, and none of these is positive semidefinite. The finite data fix the linear representative but not the positive cone, so the dimension condition remains an imported premise.
5. **The rational boundary of Everett-style branch counting.** Branch counting on fine-grained trees reproduces Born weights if and only if all weights lie in $\mathbb{Q}$. For weights in
$$\mathbb{Q}(\zeta_n)^+ \setminus \mathbb{Q}$$
, finite integer fine-graining encounters an exact arithmetic obstruction.
6. **Choice of real embedding.** The maximal totally real subfield $\mathbb{Q}(\zeta_n)^+$ has $\phi(n)/2$ real embeddings into $\mathbb{R}$ (16 at target conductor $n = 120$). Gram matrices of pure states stay positive semidefinite under every Galois conjugation, since each conjugate is again a Gram matrix, but mixed-state witnesses (starting at $n = 5$) exhibit embedding-relative positivity. The Galois group acts transitively on embeddings; selecting an ordered real embedding is an external declaration.
7. **The transcendental von Neumann entropy interface for mixed states.** For a non-pure density operator with algebraic spectrum, entropy in natural-log units,
$$S = -\sum_{\lambda_i > 0} \lambda_i \ln \lambda_i$$
, is positive and is a nonzero algebraic linear combination of logarithms of algebraic numbers. Baker's theorem on linear forms in logarithms [24,27] makes this value transcendental. The claim is unit-sensitive: in bits, $S(I_2/2) = 1$, so it does not assert that every base-two entropy is transcendental. The natural-log quantity lies outside $\mathbb{Q}(\zeta_n)$; the declared external interface uses a certified rational enclosure $[L_k, U_k]$ of width at most 2^{-k}, with its midpoint error bounded by half that width.

If finite quantum mechanics fails as a reconstruction programme, the failure is most likely to be exhibited at one of these seven boundaries. The useful outcome would be a precise theorem locating that boundary, not a change in a project status label. Historical packaging corrections and evidence-register statistics are retained in the source notes and reproduction records. They are not measures of the mathematical content established here.

5.1 Executable examples do not close the derivation gap

The repository contains executable interferometer, finite-cochain, Cayley-step, and positive-semidefinite filtering examples. They consume a declared graph, scalar field, state or amplitude data, readout rule, and experiment. Their successful execution establishes facts about those inputs; it does not derive the inputs or their quantum interpretation from primitive difference.

The exact examples retained in the main text use rational or specified algebraic arithmetic. Older numerical linear-programming examples remain numerical demonstrations. Historical execution counts and migration details are recorded in the source-notes supplement, not used as theorem counts.

6. How to read the publication package

The main paper states current mathematical results and limitations. The source-notes supplement preserves the paper-by-paper corrections, original source addresses, and detailed implementation scope. All retained source manuscripts cited there are included under `sources/original_papers/` in the paper ZIP. Companion papers used in the current exposition are included under `companion_papers/`.

The separate code-companion ZIP contains the shared Rust and Lean source trees in the same relative layout used by their manifests. This split keeps the scientific paper readable while allowing a reader to inspect and run the actual code instead of relying on absent repository paths. Section 12 gives the focused commands and distinguishes mathematical proof, finite replay, formal checking, and archive custody.

For an unresolved statement, the obligation is unchanged: provide an adequate written proof, a formal proof of the same statement, or a complete exact finite argument appropriate to its quantifiers; otherwise restrict it, identify it as open, or withdraw it. An example test remains evidence only for its stated inputs.

7. Scope limits

- Nothing here derives the Born rule, complex structure, unitarity, tensor factorization, or continuous time. Every one of those enters the source papers as a **declared import**, and importing a thing is not deriving it.
- A green test suite in this application certifies that its mechanical constraints hold — that files exist, commands exit 0, syntax parses, and unit tests execute. It does not by itself certify semantic mathematical validity or that prose claims are fully entailed by code/proof terms; those boundaries require substantive review.

8. Sources, code, and public-record corrections

The companion publication ZIP contains the retained source manuscripts under `sources/original_papers/`, the source-notes supplement, and the code used by the package-local reproductions. A separate code-companion ZIP preserves the shared Rust workspace and the Lean workspace in their repository-relative locations so the named commands do not depend on files absent from the download. Section 12 states the exact commands and the limits of the latest replay.

Two earlier Zenodo records require author-controlled correction before this edition is promoted. Prepared errata are included locally:

- `docs/zenodo_errata/20683775_erratum.md` narrows the claimed relationship between phase averaging, the Born rule, and Hilbert-space emergence;
- `docs/zenodo_errata/20550459_erratum.md` replaces the incorrect centered trace-zero dual and records the exact $K_{2,3}$ counterexample.

These files are proposed public-record corrections, not evidence that Zenodo has already been updated. The source-notes supplement contains the full version and custody discussion.

9. Limits of the proposed finite derivation

An earlier version of this section claimed a complete 50-step derivation, a minimal ambient carrier, and resolution of every proposed test. The proofs and computations do not establish those claims, so they are withdrawn. The historical wording remains available in version control.

Topic	What the Lean statements cover	What they do not cover
Carrier labels	Natural-number divisibility, a directed label system, and Gaussian record identities	An actual field colimit, all-target embedding, or minimal carrier
Probability and contextuality	Declared squared-norm weights, two Sorkin fixtures, and finite CHSH/parity arithmetic	Quotient-fiber transport, marginal no-signaling, or a derivation of Born weights
Cabello witness	Cabello matrix checks and a norm-25 integer partition	Topology or non-Bornian functions on the dense cyclotomic sphere
Discrete dynamics	Boolean endomaps, a scaled 2×2 Gaussian square root, integer Cayley identities, and integer-labelled records	Continuous Hamiltonians, logarithms, Stone's theorem, or a necessity proof
Commutator obstruction	A 2×2 integer commutator trace obstruction and one Weyl example	All dimensions and rings, a completion functor, or an exclusive continuum classification
Conductor bounds	Divisibility and least common multiples of declared labels	State embeddings, field minimality, or a general stabilizer or unextendible-product-basis theorem

The Python Cabello witness satisfies the 18-ray/9-context constraints and has a

checked negative eigenvector. The Peres computation checks 33 rays, 72 pairs and 16 triads; its binary valuation count is zero. Its finite linear system has rank 16. For the constructed function `f=1/3+z`, the residual `z`, not `f`, is orthogonal to the declared trace-free quadratic rows. Neither construction is a frame function on a dense sphere.

Python constructs ten bounded triads. Rust's legacy higher-conductor report checks four rational triads only; its former six additional offers were not computed and are withdrawn. No cross-language higher-triad parity is claimed. The topology, rational-sphere colorability, p-adic and nonstandard-model flags that lacked implementations are now explicitly unknown. Modular witnesses and four-conductor discriminant checks retain their finite scope.

Preserving an earlier prediction text does not validate its conclusion. The proposed refusals are not established, and both Gleason-related questions remain open under explicit conditions. No "probability wall is absolute" or "completion is necessary" conclusion survives this review.

The strongest result here is a collection of exact finite computations and scoped Lean theorems under declared carriers and measures. Finite-dimensional does not mean finite carrier; countable does not mean discrete topology.

10. Higher-dimensional bounds and the Tiles construction

Conductor labels here are sufficient declarations for the named objects, not a search proving minimality. In particular, a generator or parity-check matrix is not the same object as a normalized state.

- The five-qubit stabilizer generator/projector declaration uses $\mathbb{Q}(i)$, label 4; the Steane parity-check declaration uses $\mathbb{Q}$, label 1. No normalized-codeword carrier or all-stabilizer classification follows.
- The named GHZ and W normalization declarations use labels 8 and 12. A seven-qubit W normalization uses the positive root $\sqrt{7}$, discriminant 28.
- Prime-qudit normalized setting declarations combine roots of unity and the positive root $\sqrt{p}$. At $p = 7$ the bound uses 28 and $\mathrm{lcm}(120, 28) = 840$. (The field $\mathbb{Q}(\sqrt{-7})$ has discriminant -7, not 28.)
- Collins–Gisin–Linden–Massar–Popescu (CGLMP) setting bounds use

  $$\mathrm{lcm}(4d, \mathrm{disc}\,\mathbb{Q}(\sqrt{d}))$$

  ; they exclude an optimal entangled state's amplitudes. For $d = 3, 4, 5, 6, 7, 8$ the labels are respectively 12, 16, 20, 24, 28, 32, giving joint least common multiples with 120 of 120, 240, 120, 120, 840, 480.

The Tiles 3×3 example is actual matrix data, not a trace-only assertion. With five unnormalized product vectors v_i, it constructs the rational matrix

$$\rho = \frac{1}{4}\left(I_9 - \sum_i \frac{v_i v_i^{\mathsf{T}}}{v_i^{\mathsf{T}} v_i}\right).$$

Python and Rust verify mutual orthogonality, trace one, symmetry, the projector identity `4 rho^2 = rho`, and equality with its partial transpose. All 32 assignments of local orthogonality conditions are checked by exact rank-three minors, excluding a nonzero product vector in the complement. The entanglement reading uses the declared UPB-complement criterion; this is a deterministic finite certificate, not a Lean UPB theorem or general classifier. Duplicate and zero product-vector inputs are rejected.

From `python/`, `python -m finite_qm.entrypoints.replay_review_receipts --check` regenerates both repaired computation receipts byte-for-byte. Receipt identity does not promote the declarations above.

11. Principal exact results

This section collects the strongest results that emerged from the exact finite analysis. The long source-by-source record is in `finite_quantum_mechanics_source_notes.md` (`finite_quantum_mechanics_source_notes.md`). Each result below is stated by its mathematical content. Implementation names are supplied only as reproduction pointers.

11.1 Phase-blind descent requires a retained zero point

Let A be a matrix acting on nonzero vectors and let

$$\pi(c)_g = \frac{|c_g|^2}{\sum_h |c_h|^2}$$

be the phase-blind population report. A tempting claim says that $\pi(Ac)$ depends only on $\pi(c)$ exactly when every row of A has at most one nonzero entry. That claim is false on projective space. A rank-one matrix such as

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

sends every non-annihilated input to the same ray even though each row has two nonzero entries.

The row-sparsity characterization becomes correct when the quotient retains the zero vector as a distinct output. If two nonzero entries occur in one row, relative phase can be chosen so that this row cancels for one representative but not for another representative with the same population report. The retained zero point detects that annihilation. Conversely, a row-sparse matrix cannot create such phase-sensitive cross terms. The exact two-dimensional counterexamples and the zero-retaining proof are replayed in `phase_blind_quotient` and described in source-notes supplement §11.25.

11.2 Trace-zero repair quotients have a scalar-divergence dual

For Hermitian trace-zero edge defects, compare

$$\Phi_0(\delta) = \inf_{\alpha_v \in \mathrm{Herm}_0(n)} \sum_e \frac{1}{2} \|\delta_e - (d_0\alpha)_e\|_1$$

with the same infimum over all Hermitian vertex potentials. The polar of $\frac{1}{2}\|\cdot\|_1$ on $\mathrm{Herm}_0(n)$ is an observable class modulo scalar identity, with norm

$$\|[M]\|^* = \lambda_{\max}(M) - \lambda_{\min}(M).$$

Accordingly, the trace-zero dual requires spectral diameter at most one and $(d_0^* M)_v \in \mathbb{R}I$. The all-Hermitian dual instead requires $\|M_e\|_{\mathrm{op}} \le 1/2$ and $d_0^* M = 0$.

On $K_{2,3}$, orient the legs $u \to w_k \to v$ and put $D_k = 3E_{kk} - I$ on $u \to w_k$. Zero repair and the witnesses $D_k/3$ on both edges of each leg certify

$$\Phi_0 = 6.$$

The all-Hermitian potential $\alpha_u = 0$, $\alpha_v = -I$, $\alpha_{w_k} = D_k$, together with the witnesses $D_k/4$, certifies

$$\Phi_{\mathrm{all}} = \frac{9}{2}.$$

The complete theorem and dependency-free exact replay are in the companion paper *Trace-Zero Repair Quotients on a Finite Graph* [13].

11.3 CPTP monotonicity selects the trace norm on state differences

For each finite dimension n, let N_n be a norm on the real vector space

$$\mathrm{Herm}_0(n, \mathbb{C})$$

. Suppose

$$N_m(\Phi(X)) \le N_n(X)$$

for every CPTP map Φ from $M_n(\mathbb{C})$ to $M_m(\mathbb{C})$ and every

$$X \in \mathrm{Herm}_0(n, \mathbb{C})$$

. Then there is a single $c > 0$ such that $N_n(X) = c\|X\|_1$ on every nonzero such space. The dimension-one space is zero and satisfies the identity trivially.

To prove this, let $X = X_+ - X_-$ be nonzero and set

$$t = \operatorname{tr} X_+ = \|X\|_1/2 > 0$$

. For any target $Y = Y_+ - Y_-$ with $s = \operatorname{tr} Y_+ \le t$, choose an output density matrix τ and put

$$A = \frac{Y_+}{t} + \left(1 - \frac{s}{t}\right)\tau, \qquad B = \frac{Y_-}{t} + \left(1 - \frac{s}{t}\right)\tau.$$

These are normalized positive states. If P is the positive spectral support projection of X, the map

$$\Phi(T) = \mathrm{tr}(PT)A + \mathrm{tr}((I-P)T)B$$

is CPTP and satisfies $\Phi(X) = t(A-B) = Y$ exactly. Equal trace norms admit exact transports in both directions, hence the N-values are equal. Absolute homogeneity then gives the common constant c. This is a theorem on the full stated channel and matrix domain, not a selection of that domain from primitive difference or an assertion that every construction stays inside an arbitrary smaller scalar field. Exact difference-transport examples are in source-notes supplement §11.36.

11.4 Full-support Bell tables require positivity, not support alone

Four rational pair distributions can have full support while admitting no positive common joint law. The supplied qubit construction uses a rational density matrix and $(3,4,5)$-triangle observables and has

$$S_{\mathrm{CHSH}} = \frac{14}{5} > 2.$$

Every deterministic assignment is compatible with the support, so a possibilistic classifier returns no obstruction. A positive common distribution would be a convex combination of the sixteen deterministic vertices, each of which satisfies $|S| \le 2$; therefore no such distribution exists. This cleanly separates support gluing from positive probabilistic extension. The broader contextuality and no-signaling scope is in §13.8.

11.5 The Cabello frame polytope has 146 half-integral vertices

Let $v_1, \dots, v_{18}$ be the eighteen integer rays

$$\begin{gathered}(1,1,1,1),\ (1,1,1,-1),\ (1,1,0,0),\ (1,1,-1,-1),\ (1,0,1,0),\ (1,0,0,1),\\ (1,0,0,-1),\ (1,0,-1,0),\ (1,-1,1,1),\ (1,-1,0,0),\ (1,-1,-1,1),\ (1,-1,-1,-1),\\ (0,1,0,1),\ (0,1,0,0),\ (0,1,-1,0),\ (0,0,1,0),\ (0,0,1,-1),\ (0,0,0,1),\end{gathered}$$

and let the nine tetrads be their orthogonal four-element subsets. Every ray lies in exactly two tetrads, and the orthogonality graph and tetrad incidence are isomorphic to those of the Cabello configuration [2]. Put $P_i = v_i v_i^T/(v_i^T v_i)$, and let A be its 9×18 tetrad incidence matrix. The frame polytope is

$$\mathcal{P} = \{f \in \mathbb{R}^{18} : f \ge 0,\ Af = \mathbf{1}\}$$

. Its real-symmetric operator representative H_f is defined by $\mathrm{tr}(H_f P_i) = f_i$. The ten-dimensional real-symmetric projector span and the rank-nine tetrad constraints give the unique trace-one representative.

Every extreme valuation is obtained by covering the nine tetrad vertices with disjoint isolated edges of weight one and odd cycles of weight one half. An exact enumeration yields 146 valuations. Their characteristic polynomials and multiplicities are:

Multiplicity	Characteristic polynomial of H_f	Smallest eigenvalue
24	$t^2(t-3/2)(t+1/2)$	$-1/2$
72	$t^2(t^2-t-1/4)$	$(1-\sqrt{2})/2$
2	$(t-1)^2(t+1/2)^2$	$-1/2$
48	$(t^2-t/2-1/8)^2$	$(1-\sqrt{3})/4$

Hence no vertex representative is positive semidefinite; 26 have smallest eigenvalue $-1/2$, and the other 120 have smallest eigenvalue greater than $-1/4$. The complete input, enumeration argument, vertex list and exact reconstruction checks accompany the result. These statements concern the real-symmetric representatives, not arbitrary complex-Hermitian matrices that agree only on the sampled real rays. The enumeration, rank checks, and exact characteristic-polynomial certificates are described in source-notes supplement §11.52.

11.6 A single rational context pair selects $p = 2$ among power rules

Consider the positive-power family

$$P_p(i) = \frac{|c_i|^p}{\sum_j |c_j|^p}, \qquad p > 0.$$

For $\psi = (1, 1, 0)$, compare the standard basis with the rational orthonormal basis obtained by a $(3, 4, 5)$ rotation fixing the first coordinate. The shared ray has weight $1/2$ in the first context and

$$\frac{1}{1 + (3/5)^p + (4/5)^p}$$

in the second. Agreement is equivalent to $(3/5)^p + (4/5)^p = 1$. The left side is strictly decreasing for $p > 0$ and equals one at $p = 2$, so $p = 2$ is the unique positive solution. This classifies the declared power family; it is not a general Gleason theorem. Sorkin's third-order sum [7] does not make this selection: for the polynomial measures $(a+b+c)^p$ it vanishes identically at $p = 1$, the additive case, and at $p = 2$, the quadratic case (source-notes supplement §11.52, check C46). Exact rational controls for selected powers accompany the general monotonicity argument in source-notes supplement §11.58.

12. Reproduction and package boundaries

The publication is distributed as two archives with different purposes.

1. The paper archive contains this monograph, the source-notes supplement, retained source manuscripts, package-local Python reproductions, exact receipts, and checksum manifests.
2. The code-companion archive preserves the shared Rust workspace, the finite quantum mechanics native crate and campaign crate, and the Lean workspace in their repository-relative locations. Its own manifest binds every member.

The exact filename, byte count, and SHA-256 digest of the code companion are recorded in `docs/code_companion.md` and bound again by `CODE_COMPANION.sha256`.

From the extracted code-companion root, the focused trace-zero certificate is run with

```
cargo test --manifest-path omega_engine/crates_rs/Cargo.toml \
  -p omega_quantum_physics --test channels \
  trace_zero_repairs_need_uncentred_witnesses_on_two_cycles_of_qutrit_defe
↪ cts
```

The complete Lean workspace is built with

```
cd lean4/quantum_mechanics
lake build
```

The dependency-free replay for the standalone $K_{2,3}$ paper is also included at `companion_papers/trace_zero_quotient_duality_monograph.md`; its package runs with standard-library rational arithmetic and focused `pytest` tests. The paper archive is checked by

```
python scripts/regen_sha256sums.py --check
python verify_deposit.py
```

These commands have separate evidence scopes. Exact certificate replay establishes the displayed finite identities. A Rust or Lean build checks the statements actually encoded in those sources. Manifest and archive verification establish byte custody. None of these operations derives the physical premises named in the manuscript, and this edition does not claim a fresh successful build of every Rust and Lean target. The full implementation history and per-module scope are retained in the source-notes supplement.

13. The quantum carrier, observer, and measurement problem

13.1 The current answer and the questions it leaves open

Our best current answer is a conditional operational account. A quantum carrier supports the declared states, interactions and effects. An observer specifies which of those operations are available, how an apparatus registers a result, and which distinctions its reports must preserve. Measurement couples carriers and makes information available in a record. A classical description can then be sufficient for a specified set of questions even while the larger carrier retains quantum coherence.

That account separates three objects: **the probability distribution of records, the state conditioned on an obtained record, and one actual event**. Neither a report label nor a reduced density matrix supplies the last object by itself. A decision rule

evaluates a supplied record; it is not a physical mechanism that makes a particular outcome occur.

This account does not imply that every observer requires a quantum carrier. Defining an observer model requires an adequate carrier, but does not determine complex scalars, a Hilbert space, a tensor product or the Born rule. Those are currently explicit inputs to the quantum models in this paper.

There are several measurement problems, with different obligations [9]:

- **Carrier and interaction:** why this state space, composition law and measuring dynamics are physically appropriate.
- **Preferred records:** which alternatives are distinguishable and stable under the actual apparatus and environmental couplings.
- **Probabilities:** why a specified instrument has its stated outcome weights, and under which domain and assumptions those weights have a Born form.
- **Definite outcomes:** what accounts for one event rather than the joint state or its distribution of possible records.
- **Agreement:** when records from different observers can be compared and composed without assigning incompatible experiments an unsupported common history.

Decoherence addresses the suppression of accessible interference and the stability of certain records [8, 10]. It does not, under unitary evolution alone, select a single event. Our finite account makes this boundary explicit and offers tests at it; it does not announce a solution to every problem above.

13.2 The carrier and the declared observer

A finite-dimensional quantum model may have continuously many states; finite dimension, a finite presentation and a finite state set are different claims. In our exact models, entries lie in a declared field such as the Gaussian rationals or a cyclotomic extension, with a specified conjugation and real embedding. State admission requires Hermiticity, positivity and trace one. Effects satisfy the operator inequalities between zero and the identity. Some spectral operations require scalar extensions or supplied witnesses; an exact implementation can refuse an unsupported input.

One abstract observer model,

$$\mathcal{O} = (K, \mathcal{F}, \mathcal{R}, \delta, d_0, d_1, J, \mathcal{M}, \tau),$$

names the carrier, local spaces, comparisons, defects, repairs, closure test, gauge, compatible transformations and decision rule. For a quantum experiment it also needs a preparation domain, system/apparatus/environment split, interaction, instrument, accessible effect menu, readout and calibration conditions. A physical apparatus realizes part of that model; a person or conscious experience is not a required mathematical component.

One declared obstruction observer uses Hermitian edge defects and

$$J_Q(h) = \frac{1}{2}\sum_e \|h_e\|_1, \qquad \Phi_Q([\delta]) = \inf_\alpha J_Q(\delta - d_0\alpha).$$

For a finite-dimensional real space of Hermitian edge defects and a linear repair image, this is the quotient norm induced by J_Q; finite dimension makes that image closed. It does not identify every formal repair with a realizable quantum operation. Nor is every algebraic projector a positive or completely positive channel. Positivity, physical availability and question preservation must be checked separately.

Observer relativity therefore means that the experiment and evaluation are specified. It does not mean that an arbitrary decision label changes the physical state, or that different descriptions license contradictory reports about the same compatible experiment.

13.3 Interaction, readout and conditional updating

On a supplied finite quantum carrier, an instrument has completely positive maps indexed by record labels. In Kraus form,

$$\mathcal{J}_i(\rho) = \sum_a K_{ia}\rho K_{ia}^\dagger, \qquad \sum_{i,a} K_{ia}^\dagger K_{ia} = I.$$

The sum of these maps preserves trace. Given the positive trace pairing,

$$E_i = \sum_a K_{ia}^\dagger K_{ia}, \qquad p_i = \operatorname{tr}(E_i\rho) = \operatorname{tr}\mathcal{J}_i(\rho), \qquad \rho_i = \mathcal{J}_i(\rho)/p_i \quad (p_i > 0).$$

The unnormalized maps are linear; normalization is conditional on a nonzero-probability record. A zero-probability branch has no normalized conditional state. The non-selective state is the sum of the unnormalized branches. These identities follow from the declared instrument and pairing; they are not a reconstruction of either from inscription histories.

For an ideal projective instrument, the Kraus operators are the orthogonal projectors themselves. This gives the Lüders conditional update and an idempotent non-selective dephasing channel. General instruments need not be projective, repeatable or idempotent. An effect specifies an outcome's probability functional, but does not uniquely specify its disturbance map. The measuring interaction and record grouping are therefore necessary data.

The exact finite implementation checks these identities and a dilation for admitted Gaussian-rational inputs. It does not generate one physical event. Conditional updating answers what follows **if this record occurred**; it cannot be used as its own explanation of occurrence.

13.4 An exact record with recoverable coherence

Take two qubits, a system and an apparatus, with

$$|\psi\rangle = \tfrac{3}{5}|0\rangle + \tfrac{4}{5}|1\rangle, \qquad |A\rangle = |0\rangle.$$

A controlled-NOT from system to apparatus gives

$$|\Psi\rangle = \tfrac{3}{5}|00\rangle + \tfrac{4}{5}|11\rangle, \qquad \rho_{SA} = |\Psi\rangle\langle\Psi|.$$

Compare this with the dephased joint state

$$\sigma_{SA} = \tfrac{9}{25}|00\rangle\langle 00| + \tfrac{16}{25}|11\rangle\langle 11|.$$

Both states give the same system reduction, apparatus reduction and pointer distribution: the diagonal weights are nine twenty-fifths and sixteen twenty-fifths. Repeated sharp pointer reads are consistent within the ideal instrument. This is an available classical record distribution under the declared pointer readout. It is not a proof that the joint state is the mixture or that one pointer label was selected.

Their difference has only two nonzero entries, each twelve twenty-fifths, between the basis vectors labelled 00 and 11. Its eigenvalues are

$$\tfrac{12}{25}, \quad -\tfrac{12}{25}, \quad 0, \quad 0, \qquad D_{\mathrm{tr}}(\rho_{SA}, \sigma_{SA}) = \tfrac{12}{25}.$$

The bounded observable consisting of Pauli X on each qubit distinguishes them: its expectation is twenty-four twenty-fifths for the coherent state and zero for the mixture. Alternatively, applying the same controlled-NOT again recovers the initial coherent system state from the first preparation; the mixture yields a diagonal system state with the apparatus reset to zero. The system's Pauli-X readout then distinguishes them by the same amount.

Every matrix entry in this example is rational. Its conclusion depends on the admitted joint controls. If the apparatus or environment cannot be controlled coherently, this erasure experiment may be unavailable. Extending the model with an orthogonal environmental record can remove the coherence from the system/apparatus reduction while retaining it globally. Losing access to that distinction explains the adequacy of a reduced description for a restricted task; it is not a derivation of objective collapse.

13.5 When a classical report is sufficient

Let the report be a map from the admissible histories or states to a record space, and let a protected question have answer map q. Exact sufficiency is

$$q = \hat{q} \circ p \quad \Longleftrightarrow \quad p(x) = p(y) \implies q(x) = q(y).$$

The factor is unique on the report image. Equality of the two maps' fibers would impose a stronger faithfulness condition; factorization only requires that the report never identifies different protected answers. For a future experiment, the answer includes its probabilities or conditional reports, so present agreement alone may be insufficient.

For a supplied nonempty effect menu, denoted separately from the compatible transformation class by $\mathcal{E}_{\mathrm{adm}}$, define accessible distinguishability by

$$D_{\mathcal{E}_{\mathrm{adm}}}(\rho, \sigma) = \sup_{E \in \mathcal{E}_{\mathrm{adm}}} |\operatorname{tr}(E(\rho - \sigma))|.$$

It is bounded by trace distance. The full complex effect domain implies equality; a restricted menu also attains equality if it contains an optimal distinguishing effect for the given pair. An arbitrary finite menu need not. A classical description is adequate at a declared tolerance only when it preserves the protected answers under the

admitted preparations, future controls, readouts and time horizon. If a later control converts a discarded phase into a readable population, the original population report was not dynamically sufficient. An algebraic completion may require observables the apparatus cannot measure jointly or cannot access at all.

A sharp pointer record algebra generated by orthogonal projectors is commutative. The complete quantum algebra containing it need not be; even the block-diagonal fixed-point algebra of dephasing can retain noncommuting degrees of freedom within a higher-rank block. Thus a classical record can be a subalgebra of a quantum carrier, rather than a moment at which the entire carrier becomes classical. Its preferred basis comes from a specified coupling and stability criterion. A degenerate coupling does not choose a unique basis without further apparatus information.

The existing finite sufficient-record derivation (`research/quantum_to_classical_descent/README.md`) establishes static factorization and invariant-row-space completion on its declared linear carriers. It also distinguishes autonomous expectation coordinates from jointly sharp measurements and from a realized event. Its finite certificates and conditional analytic bounds retain those premises.

13.6 Outcomes, observer agreement and the remaining boundary

There is a useful elementary restriction on a density-only deterministic outcome rule. Suppose an affine map sends every density state to a vertex of a classical probability simplex and sends two pointer eigenstates to distinct vertices. On their equal mixture, affinity requires the midpoint of those vertices, which is not a vertex. No such map satisfies all three requirements. This argument assumes the density state is the complete input and classical mixing is represented affinely. It does not exclude an additional configuration, a stochastic event law, a relative-branch account or another explicitly enlarged carrier.

Different interpretations provide different further premises. A definite configuration account supplies variables and dynamics beyond the density description; an objective-collapse account modifies dynamics; a relative-state account changes how outcomes are associated with observers and branches; an operational account can take outcome occurrence as a primitive. These are distinct positions, not alternative names for the same derived theorem. None follows solely by assigning a software label or discarding an environment.

Records from multiple observers require compatible instruments, a declared comparison protocol and retained history. Separate outcome tables do not automatically admit a common nonnegative joint assignment across incompatible contexts. Conversely, contextuality does not prevent agreement on compatible pointer records. An observer who performs a coherent erasure has changed the experiment relative to one who preserves and rereads the mark; a comparison must include that change rather than merge both into an undeclared history.

Our current answer therefore explains, under supplied quantum premises, how interactions furnish record probabilities, why particular reduced descriptions can be adequate and stable, and how conditional updates compose. It leaves two foundational burdens explicit: deriving the operational carrier from native histories, and

accounting for actual outcome occurrence under a stated ontology or event law. Both questions remain open.

13.7 Testing two foundational carrier types

Begin with the hypothesis that a quantum carrier and a classical carrier are distinct foundational types. The claim becomes testable only after declaring their state spaces, readouts, allowed transfers and composition. Calling the types distinct does not specify a prediction. Conversely, embedding a classical register into a quantum matrix algebra does not prove that only quantum things exist. We separate the following versions of the hypothesis.

Hypothesis	Finite operational content	Result here
Two types coexist	Quantum systems and classical registers have their own admitted states and transfers.	Consistent hybrid models exist; the bare claim is not refuted.
A population record is the complete future carrier	Presently equal populations imply equal answers after every admitted control.	Refuted on the declared coherent-control protocol below.
Classical storage faithfully replaces quantum storage	Positive, normalized encoding and retrieval preserve arbitrary unknown qubit inputs.	Refuted by the sharp four-state bound below.
Additional classical variables explain quantum predictions	An enlarged classical description may depend on preparation, context, hidden variables or nonlocal dynamics.	Not excluded by the storage bound; its own premises and predictions must be tested.
The full quantum carrier is all that exists	No additional ontology is needed or exists.	Operational sufficiency does not decide this ontological assertion.

A consistent two-type countermodel. Let

$$\mathcal{A} = \bigoplus_{\lambda=1}^{m} M_2(\mathbb{C}).$$

A state is a tuple $(\rho_\lambda)_\lambda$, with each block positive and

$$\sum_\lambda \operatorname{tr} \rho_\lambda = 1.$$

Effects are tuples $0 \le E_\lambda \le I$; their probabilities are

$$\sum_\lambda \operatorname{tr}(E_\lambda \rho_\lambda).$$

The center carries a classical label, while each block retains noncommuting qubit observables. Stochastic label changes and within-block quantum channels furnish consistent finite operations. A normalized POVM and supplied density preparations give the block transfer

$$\rho_\lambda = \operatorname{tr}(F_\lambda \rho)\sigma_\lambda.$$

Summing the blocks discards the label. The model can be presented over $\mathbb{Q}(i)$ on its exact finite fixtures.

This is a genuine restriction of the full M_{2m} quantum model if coherences and controls between label blocks are prohibited. Its matrix embedding does not remove that restriction or decide its ontology. It also supplies a distribution on labels, not a derivation of one occurring event. Thus the existence of two operational types is mathematically consistent. A claim that a particular apparatus must inhabit such a restricted type needs evidence for the prohibited coherences and controls.

Present equality is not future equality. In the declared qubit basis let

$$|\psi_\pm\rangle = \tfrac{3}{5}|0\rangle \pm \tfrac{4}{5}|1\rangle, \qquad U = \begin{pmatrix} 3/5 & 4/5 \\ -4/5 & 3/5 \end{pmatrix}.$$

Both states have population report $(9/25, 16/25)$, but after the same available unitary U, the probability of the first basis outcome is respectively 1 and $49/625$. A model that identifies these states as the same complete state cannot predict both answers using only that record. Dephasing them to the same diagonal density operator makes their future answers equal; it has changed the state, rather than faithfully retained the discarded phase. This establishes that the population report is insufficient for that future question. It does not rule out classical storage of additional variables, nor assume that the displayed control is available in every device.

A sharp classical-storage obstruction. Consider an unconditional, trace-preserving transfer of an unknown input qubit through a finite classical alphabet and back to a qubit. The intermediate outcome may be random; the protocol must succeed without postselection. Use the full positive qubit state/effect model and affine mixing. The encoder sees only the input state; it has no preparation label, quantum side memory or retained coherent environment. The decoder returns a normalized positive qubit state for each letter. No postselection or conditional success loophole is included. The test basis is disclosed only after the returned qubit is fixed; requesting a basis-dependent classical answer is a different protocol. These premises give the measure-and-prepare form

$$\Phi(\rho) = \sum_\lambda \operatorname{tr}(F_\lambda \rho)\sigma_\lambda, \quad F_\lambda \geq 0, \quad \sum_\lambda F_\lambda = I, \quad \sigma_\lambda \geq 0, \quad \operatorname{tr}\sigma_\lambda = 1.$$

A positive affine encoder on the full density-state domain has POVM response operators; an affine decoder from a probability simplex is specified by its states at the vertices. Thus the displayed form follows from the stated operational premises. It is also the standard classical-transfer form of an entanglement-breaking channel [11]. The finite Choi matrix, with input factor first, is

$$J_\Phi = \sum_\lambda F_\lambda^T \otimes \sigma_\lambda.$$

It is positive, and tracing its output factor gives I; complete positivity and trace preservation therefore follow without choosing Kraus square roots outside the coefficient field.

Let the four equiprobable inputs be

$$P_{Z\pm} = \tfrac{1}{2}(I \pm Z), \qquad P_{X\pm} = \tfrac{1}{2}(I \pm X),$$

and define their average survival probability

$$S(\Phi) = \frac{1}{4} \sum_{a\in\{X,Z\},\, s\in\{+,-\}} \operatorname{tr}(P_{as}\Phi(P_{as})).$$

All four projectors have rational entries. The exact theorem is

$$\boxed{S(\Phi) \le \frac{3}{4}.}$$

It holds for every finite number of classical letters under the premises above, and the constant is attained.

For a proof, use the real trace coordinates

$$w_\lambda = \operatorname{tr} F_\lambda, \qquad f_{\lambda a} = \operatorname{tr}(F_\lambda a), \qquad s_{\lambda a} = \operatorname{tr}(\sigma_\lambda a).$$

The two-by-two determinant and trace conditions for positivity imply

$$f_{\lambda X}^2 + f_{\lambda Z}^2 \le w_\lambda^2, \qquad s_{\lambda X}^2 + s_{\lambda Z}^2 \le 1.$$

The omitted Y coordinates can only strengthen these inequalities. Cauchy–Schwarz and $w_\lambda \ge 0$ give the nonnegative margin

$$d_\lambda = w_\lambda - f_{\lambda X}s_{\lambda X} - f_{\lambda Z}s_{\lambda Z} \ge 0.$$

Expanding the four traces and using $\sum_\lambda w_\lambda = 2$ yields

$$S(\Phi) = \frac{1}{2} + \frac{1}{8}\sum_\lambda (f_{\lambda X}s_{\lambda X} + f_{\lambda Z}s_{\lambda Z}).$$

$$\frac{3}{4} - S(\Phi) = \frac{1}{8}\sum_\lambda d_\lambda.$$

Measuring Z and preparing its obtained eigenstate gives survival probabilities $(1, 1, 1/2, 1/2)$ in the order $Z+, Z-, X+, X-$, attaining $S = 3/4$. Identity quantum storage gives $S = 1$. A faithful positive classical replacement preserving even these four preparations and their retrieval tests therefore cannot exist under these premises. More classical letters, shared randomness independent of the input, or an additional letter cannot improve the bound: each normalized strategy has the same inequality, and convex mixtures retain it.

For a declared error criterion, if all four survival probabilities are at least $1 - \varepsilon$, the bound requires $\varepsilon \ge 1/4$. More generally, if each retrieved state is within trace distance ε of its input, its own projector's survival probability is at least $1 - \varepsilon$, giving the same

necessary bound. Randomly choosing between X and Z measurement/preparation with equal weights attains survival $3/4$ and trace distance $1/4$ on each of the four states: the retrieved state is $\frac{3}{4}P + \frac{1}{4}(I - P)$, whose difference from P has eigenvalues $\pm 1/4$. Both four-state error bounds are sharp. This is a bound on an unconditional classical-storage protocol, not a noise estimate for an arbitrary apparatus or a statement about selectively successful trials.

A static classical control and the boundary of the conclusion. Four classical letters with predetermined signs $(z, x) \in \{+, -\}^2$ reproduce all the four preparations' static X/Z readouts: for a Z eigenstate, put equal probability on the two matching z letters, and similarly for an X eigenstate. This also preserves the equal-mixture identity

$$\tfrac{1}{2}(P_{Z+} + P_{Z-}) = \tfrac{1}{2}(P_{X+} + P_{X-}) = I/2.$$

These finite tables do not refute a classical explanation.

What they cannot supply is a positive encoding/retrieval on the full qubit carrier. A letter with definite $x = z = +1$ would require a retrieved qubit with both expectations equal to one: the unit-probability readouts force that requirement on every letter with nonzero weight in both preparations. The candidate $(I + X + Z)/2$ has determinant $-1/4$; allowing an extra Y coefficient makes this determinant still smaller. Similarly, extending that letter's response probability affinely to all qubits requires $F_{++} = (I + X + Z + bY)/4$, whose determinant is $-(1 + b^2)/16$. Positivity fails. Restricting the admitted states and probes to the four-state convex hull, allowing contextual responses, or supplying a preparation label changes these premises and can evade the conclusion. A classical computer storing known matrix entries is also a description of inputs, not a positive physical encoding of an unknown qubit.

The result establishes a concrete asymmetry: classical records can live inside a supplied quantum model, and may be sufficient for selected questions; an unknown quantum state cannot in general be replaced and restored through a solely classical positive carrier. Classical-looking behavior can reflect restricted access to coherence, actual loss of coherence, or a restricted hybrid model. Those possibilities must be distinguished by declared future controls and protected questions. Neither this obstruction nor coherent erasure refutes every two-type ontology, establishes that quantum ontology is exhaustive, or supplies the missing event-selection law.

13.8 Three separate carrier distinctions

The two-type hypothesis becomes sharper when three different obstructions are kept separate. A report can lose a required future answer; compatible local probability tables can fail to have a positive global extension; and a declared classical center can conflict with purification of every state. None of these statements alone establishes that the quantum carrier is all that exists.

Test 1: future-question sufficiency. Supply a finite preparation family, a present POVM report, a CPTP control and a future POVM. The report is the whole ensemble probability vector, not one sampled outcome. Ask whether the probability of outcome zero of the future POVM factors through that report. Two preparations with the same report and different answers disprove such a decoder; otherwise a finite decoder is

returned on the supplied family. For

$$|\psi_\pm\rangle = (3|0\rangle \pm 4|1\rangle)/5, \qquad U = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix},$$

Z populations coincide at $(9/25, 16/25)$, but the probabilities of zero after U are 1 and $49/625$. Identity control passes the same sufficiency test. Reporting the pulled-back future effect also restores sufficiency on this family. This is the finite protected-question criterion and the phase-blind descent obstruction on an exact fixture. A pass on these two states does not prove sufficiency for every preparation or every future control.

Test 2: positive Bell gluing. Supply the four binary pair distributions for settings $00, 01, 10, 11$. Ask for nonnegative weights on the sixteen assignments (a_0, a_1, b_0, b_1) reproducing all four marginals. Exact convex membership returns either those weights or a separating functional verified on all sixteen deterministic vertices. Signed solutions to linear equations are insufficient. For $|\Phi^+\rangle$, take

$$A_0 = Z, \quad A_1 = X, \quad B_0 = (4Z + 3X)/5, \quad B_1 = (4Z - 3X)/5.$$

The correlation vector is $(4/5, 4/5, 3/5, -3/5)$, hence

$$S = E_{00} + E_{01} + E_{10} - E_{11} = 14/5 > 2$$

. Every context probability is strictly positive: their numerators over 20 are respectively

$$(9, 1, 1, 9), (9, 1, 1, 9), (8, 2, 2, 8), (2, 8, 8, 2)$$

, in $(++, +-, -+, --)$ order. Thus a support-only compatibility test admits all sixteen assignments and misses this obstruction. Another regression has $S = 0$ for this orientation but violates a different CHSH facet; the implementation tests the complete convex hull. Uniform independent outcomes provide the positive extension control. The numerators displayed here are the ones encoded in the Lean fixture; the Rust violation door V2 in source-notes supplement §11.33 uses the companion orientation $B_1 = (3Z - 4X)/5$, with correlation vector $(4/5, 3/5, 3/5, -4/5)$ and the same $S = 14/5$.

Bell locality and setting independence explain why one would require such a joint table in a physical model; they are additional premises, not outputs of the solver. In detail, suppose a common, setting-independent probability measure μ on hidden states λ and local response distributions $A(a \mid x, \lambda)$, $B(b \mid y, \lambda)$ give the four pair tables. Then

$$w(a_0, a_1, b_0, b_1) = \int A(a_0 \mid 0, \lambda) A(a_1 \mid 1, \lambda) B(b_0 \mid 0, \lambda) B(b_1 \mid 1, \lambda)\, d\mu(\lambda)$$

is a nonnegative normalized table on the sixteen assignments and has those pair marginals: sum out the unused response factors, whose sums are one. This reduction works for finite, countable or continuous hidden-state spaces, provided the stated probabilities are measurable. Increasing their cardinality does not evade the obstruction. This finite result excludes that conjunction for the supplied statistics. It does not exclude nonlocal hidden-variable models or establish a unique ontology. The obstruction is failure of positivity-constrained global gluing, which must not be conflated with signed linear exactness or with a support-only contextuality obstruction. The source manuscript on finite Bell bounds (source-notes supplement §11.33) already identifies

this joint-table premise; this test adds a shared executable experiment [26] and a Lean marginal-to-CHSH proof.

Test 3: central sectors and purification. Declare the full positive normalized states of $A = \bigoplus_i M_{n_i}(\mathbb{C})$, standard tensor composition with another such algebra, and the available ancilla block sizes. A state has normalized central weights p_i and block states ρ_i. A pure composite state occupies one joint central block, so its marginal has one central weight equal to one. A state with two positive central weights therefore has no pure extension within this model, for any ancilla of this form. More quantitatively, every extension ω of that state and every marginal σ of a pure composite state obey

$$\mathrm{Tr}(\omega^2) \le \sum_i p_i^2, \qquad \tfrac{1}{2}\|\rho - \sigma\|_1 \ge 1 - \max_i p_i.$$

For the first inequality, sum the squared weights of normalized joint blocks, use purity at most one within each block, and then use

$$\sum_j p_{ij}^2 \le (\sum_j p_{ij})^2 = p_i^2$$

. For the second, the central measurement is a trace-distance contraction and distinguishes p from a point mass by $1 - p_k$. With a sufficiently large quantum ancilla both bounds are sharp: purify each ρ_i inside a chosen ancilla block for the first; for the second purify ρ_k in a block of maximal central weight. The half-half classical mixture gives both bounds $1/2$. The exact finite calculation separately computes tighter purity, rank and marginal-distance constraints for the listed ancilla sizes.

A single sector has no central obstruction, but an undersized ancilla can still prevent purification. The control with spectrum $(9/25, 16/25)$ needs an ancilla block of size at least two. Requiring purification for every state rules out nontrivial centers only under these state-space, composition and purity premises. It neither rules out a consistent hybrid model that declines that requirement nor derives purification as a law of nature. This is the conditional center-elimination step in the companion obstruction analysis; real quantum theory's purification and the associativity of quaternion multiplication are unaffected.

13.8.1 Relation to finite report quotients The three tests address distinct interfaces. Phase-sensitive readouts ask whether a report preserves answers to declared future experiments. Positive Bell extension asks whether compatible pair distributions are marginals of one nonnegative joint law. The central-sector bounds ask what extensions are possible in a specified direct-sum tensor model. None of these questions selects the underlying quantum state space.

A finite protected-question quotient identifies two histories only when every admitted continuation gives the same answer and remains available. That is a minimal report relative to one finite experiment, not a derivation of positive operator algebras, complex scalars, or tensor composition. Similarly, true marginals and products of one-site marginals can each satisfy their consistency conditions even when their difference is nonzero. Report locality, local tomography, and Bell local causality therefore require different hypotheses. The detailed comparison and local source addresses are in the source-notes supplement and companion manuscript [12].

13.8.2 From primitive difference to Bell: the endpoint and the open step Two questions must be separated. First, the finite algebra of a Bell/CHSH model closes once its state, observables, composition and probability rule are supplied. Second, primitive difference alone does not force those quantum inputs. Positive L^1 weights are additional carrier data, not a theorem of a bare distinction relation. Once four binary response variables

$$A_0, A_1, B_0, B_1 \in \{-1, +1\}$$

share one positive normalized joint law, every deterministic assignment has

$$S = A_0(B_0 + B_1) + A_1(B_0 - B_1) \in \{-2, 2\},$$

and convexity gives $|S| \leq 2$. This conditional joint-extension theorem is proved in source-notes supplement §11.33; it is not a derivation of Bell's theorem from L^1 alone. In the CHSH scenario, Fine's theorem identifies existence of the relevant common joint distribution with satisfaction of the complete Bell/CHSH family under the compatibility assumptions [16]. Bell's physical conclusion additionally uses locality and an identification of preparations, settings and outcomes [14,15].

The quantum side is now equally explicit. On the declared bipartite carrier, let

$$\begin{aligned} A_0 &= Z, & A_1 &= X, \\ B_0 &= (Z + X)/\sqrt{2}, & B_1 &= (Z - X)/\sqrt{2}, \end{aligned}$$

and use the entangled state

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

with the trace/Born pairing. The exact implementation constructs the CHSH operator and verifies the two sum-of-squares inequalities

$$-2\sqrt{2}I \leq B_{\mathrm{CHSH}} \leq 2\sqrt{2}I\,.$$

The same witness evaluates to $S = 2\sqrt{2}$ [17]. Every matrix entry in this witness is fixed by complex conjugation: the construction lives in the real field $\mathbb{Q}(\sqrt{2})$. It is also static. Therefore a specifically complex $U(1)$ phase fiber and continuous unitary time evolution are not necessary for a Tsirelson-saturating CHSH witness.

Boundary	What is now closed	What remains supplied or open
primitive difference to positive weights	finite difference/cochain and quotient machinery	positivity and normalized preparation weights are admitted data
classical Bell face	one positive global joint law implies $\|S\| \leq 2$ exactly	physical locality, setting independence and experiment identification

Boundary	What is now closed	What remains supplied or open
strict quantum violation	rational density matrix and observables give $S = 14/5 > 2$ and no positive joint table	tensor composition, entangled preparation, measurements and Born/ trace readout
quantum endpoint	exact SOS upper bound and exact witness give $S = 2\sqrt{2}$ over $\mathbb{Q}(\sqrt{2})$	derivation or operational selection of the quantum inputs from native histories

This also corrects the proposed "five-import boundary." Its five imports (the phase fiber, the quadratic readout, context parity, the frame function and the clock tower, each mapped to finite data in a research note `research/qm_207_*` to `research/qm_211_*`) remain useful as a catalogue of structures used by **full complex quantum mechanics**, but they are neither proved independent nor a minimal basis for Bell. Quadratic/inner-product geometry and the Born pairing overlap rather than forming an established independence theorem. Same-party noncommutativity is load-bearing for the maximal CHSH construction, while cross-party observables commute. Bipartite composition and entangled preparation are load-bearing but absent from the five-item cascade. Complex phase and continuous dynamics belong to the larger theory, not to the static Bell minimum.

Verdict. The finite algebraic road ends at Bell: conditional on the declared quantum interface, this repository now contains the classical bound, an exact violation, the Tsirelson upper certificate and an attaining witness. The foundational road from primitive difference does not end there. Its remaining step is the first derivation target in Source Notes §6 (retained subsection §13.9): derive or operationally justify the positive composite algebra, state, effects and pairing without silently importing the quantum model that the Bell calculation uses.

13.8.3 Čech contextuality and the Regge firewall The sheaf-theoretic language sharpens the local-to-global part of this story, but only when its objects and degrees are kept distinct. A measurement scenario consists of a measurement set and a cover by compatible contexts. An empirical model is a family of probability distributions on those contexts whose marginals agree on overlaps. In Bell scenarios this compatibility is the no-signalling condition; in general contextuality scenarios it is often called no-disturbance. This is the compatible-family, or Čech degree-zero, condition for the declared distribution presheaf [18]. It is not yet a first-cohomology obstruction.

Abramsky and Brandenburger identify a global section of the distribution presheaf with a noncontextual hidden-variable realization for a general measurement cover [18]. Fine's theorem is the more specific CHSH result used in §13.8.2 [16]. The cohomological refinement instead chooses a local section in an abelian presheaf derived from the support and constructs an obstruction class. A nonzero class is sufficient for contextuality, but the test is not complete: contextual models with a vanishing chosen cohomological obstruction exist [19]. Three distinctions follow.

1. Vanishing of the relevant obstruction class does not say that the whole cohomology group is zero.
2. Compatible local distributions do not automatically constitute a nonzero Čech H^1 class; the coefficient presheaf, chosen local section and coboundary maps must be declared.
3. The scalar CHSH excess

$$\Delta_{\mathrm{CHSH}} = \max(0, |S| - 2)$$

is a useful violation margin, not a magnitude or norm of a Čech class. Thus the rational witness in §13.8.2 has $\Delta_{\mathrm{CHSH}} = 4/5$, and a PR box would have $\Delta_{\mathrm{CHSH}} = 2$, but these arithmetic facts do not define a cohomology theory. The rational witness excludes a positive global joint law; it does not, without a separately declared support construction, compute a Čech obstruction. PR boxes are among the examples for which the standard support-derived obstruction is known to be nonzero [19].

The Mermin–Peres scenario does support an exact finite local-to-global obstruction. Let

$$d_0 : \mathbb{F}_2^V \to \mathbb{F}_2^C$$

map nine observable values to the six context parities, let $s \in \mathbb{F}_2^C$ be the declared sign vector, and let ℓ sum all six context coordinates. Every observable occurs twice, so $\ell d_0 = 0$, while the odd target has $\ell(s) = 1$. Hence $s \notin \operatorname{im} d_0$. Every individual context and every five-context subfamily is satisfiable, but the complete six-context family is not. This is the exact locally satisfiable but globally inconsistent parity obstruction verified in source-notes supplement §11.50 and is the finite combinatorial core of the Peres–Mermin argument [5,6,20]. It is safe to call it a nonzero cokernel class. Calling it an H^1 class additionally requires a declared next coboundary d_1 with $d_1 s = 0$; calling it a class in H^2 requires a declared two-complex. The bare carrier graph has no degree-two cochains. Finally, the parity contradiction itself is finite $\mathbb{F}_2$ algebra, but identifying its signs with quantum measurements uses the supplied commuting two-qubit Pauli operator realization. It is not a derivation of that quantum realization from parity alone.

Phase holonomy needs the same firewall. Transition functions for a specified $U(1)$ bundle can form a Čech 1-cocycle, but Aharonov–Bohm and Berry holonomies also depend on a connection and a loop [22,23]. Nontrivial connection holonomy can occur on a trivial bundle, so it is not automatically a nonzero class in a bare $\check{H}^1$ of transition data. Likewise, calling a phase a coboundary requires a declared coefficient sheaf, cover and coboundary map. The paper therefore retains phase holonomy as a conditional structural analogy, not as a general identification with the contextuality obstruction above.

The same discipline blocks the proposed Regge identification. Regge calculus places curvature on codimension-two hinges, with deficit

$$\epsilon_h = 2\pi - \sum_{\sigma \supset h} \theta_{\sigma h}$$

for the appropriate dihedral angles [21]. In a two-dimensional equilateral triangulation the hinge is a vertex: six triangles give zero deficit, five a positive deficit and seven a negative deficit. That is a geometric curvature diagnostic. It neither supplies a measurement cover nor empirical probabilities. Even zero local deficits do not, without global hypotheses, force trivial topology or holonomy.

Accordingly, no result here identifies a Čech obstruction with $|\epsilon_h|$, establishes the bound

$$S_{\mathrm{CHSH}} \leq \min(4, 2 + |\epsilon_h|)$$

or makes curvature a prerequisite for Bell violation. The existing experimental API records the angle but returns `None` for the Čech obstruction, Bell bound and contextuality flag, including at zero deficit. The exact CHSH witness in §13.8.2 is independent of Regge curvature and reaches $2\sqrt{2}$, so the proposed unconditional implication $\epsilon = 0 \Rightarrow |S| \leq 2$ cannot be a general quantum theorem.

Narrow claim and next test. Čech cohomology can witness a declared local-to-global obstruction, while Regge deficits measure declared simplicial curvature; their resemblance is presently an analogy, not a bridge. A genuine bridge must map a specified triangulation and hinge data to a measurement cover, outcome presheaf and empirical distributions, prove overlap compatibility, define the coefficient complex, and then derive a bound that survives both a classical control and the zero-deficit quantum CHSH counterexample. Until those data and proofs exist, the proposed bridge remains open.

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